AWANTIKA SRIVASTAVA

Data Scientist | Machine Learning Engineer | NLP & RAG

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PROFILE SUMMARY

Data Scientist with **2+ years of experience in machine learning, statistical analysis, and data-driven problem solving**. Skilled in building **classification and regression models using Python, Pandas, NumPy, and Scikit-learn** on large datasets. Experienced in **data cleaning, preprocessing, feature engineering, and exploratory data analysis (EDA)** to extract meaningful insights. Hands-on experience in developing **end-to-end ML pipelines, model evaluation using cross-validation techniques, and monitoring model performance in production systems**. Passionate about applying machine learning to **fraud detection, risk prediction, and financial analytics problems**.

CORE TECHNICAL SKILLS

- **Programming & Data Analysis:** Python, SQL, Numpy, Pandas, Matplotlib, Seaborn
- **Statistics & Mathematics:** Statistical Modeling, Hypothesis Testing, A/B Testing, Probability, Linear Optimization, Trend Analysis.
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, Hyperparameter Tuning, Feature Engineering, EDA, Data Cleaning, Data Preprocessing
- **Deep Learning & AI:** Neural Networks, CNN, RNN, LSTM, Transformers (BERT), Transfer Learning, Model Fine-tuning, RAG, LLMs, TensorFlow, PyTorch
- **Model Evaluation:** Cross Validation, Feature Importance, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix
- **Computer Vision**: Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Video Analytics, OpenCV, Paddle OCR
- **NLP & GenAI:** Text preprocessing, Tokenization, Chunkins, Transformers, Sentiment Analysis, RAG, LLM Applications, Semantic Search, HuggingFace, LangChain, FAISS
- **MLOps & Deployment:** Flask, REST APIs, Model Serving, Git, Docker (basic), ML Pipelines
- **Cloud & Platforms:** AWS (EC2, S3 – basic), Linux

EXPERIENCE

Data Scientist | PPS International Pvt. Ltd. **January 2024-Present**

- Developed **machine learning models for classification and regression tasks** using Python and Scikit-learn to support data-driven decision making.
- Performed **exploratory data analysis (EDA), data cleaning, and preprocessing** on large structured datasets using Pandas and NumPy.
- Implemented **feature engineering and feature selection techniques** to improve predictive model performance and model stability.
- Applied **cross-validation, hyperparameter tuning, and model evaluation metrics (Precision, Recall, F1-Score, ROC-AUC)** to ensure model robustness and reduce overfitting.
- Built **automated ML pipelines for data preprocessing, feature engineering, model training, and evaluation** to streamline experimentation and improve development efficiency.
- Created **data visualization dashboards using Matplotlib and Seaborn** to communicate insights and model performance to stakeholders.
- Developed **REST APIs using Flask** to serve machine learning models for real-time and batch predictions.
- Monitored **model performance and prediction distributions** to detect potential **data drift and model degradation** in production environments.
- Collaborated with **data scientists, engineers, and product teams** to translate business problems into machine learning solutions.

PROJECTS

Railway Driver Assistance System (RDAS) | Enterprise ML Project

- Designed and deployed a **real-time ML systems** for unsafe behavior detection using **CNN-based object detection** models.
- Implemented **end-to-end ML pipelines** from data ingestion to production inference.
- **Deployed** optimized models using **TensorFlow Lite** for continuous, **low-latency inference.**
- Developed a **Flask-based** web interface to display detections and recorded video clips in **real time**.
- **Deployed** the complete solution on **edge devices**, enabling reliable on-device inference and **real-time event recordings**.

Amazon Stock Price prediction | Applied ML Project

- **Built batch-oriented ML workflows** for time-series forecasting using **LSTM** models.
- Implemented data **preprocessing**, feature engineering, and model **evaluation pipelines** on **large** historical datasets.
- Designed **sliding-window** based **sequence generation** and trained LSTM model 5-year stock price data.
- Evaluated model performance using appropriate regression **metrics** and **trend-based analysis** for short-term forecasting.

YouTube Comments Sentiment Analyzer | link -https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/

- Deployed transformer-based **NLP models** as **production-ready** services with **REST APIs**.
- **Fine-tuned** and served a **DistilBERT**-based **sentiment** classification model for large-scale text inference.
- **Built** and deployed an interactive **streamlit web application** to perform real-time **sentiment analysis** on YouTube comments.
- **Processed** high-volume text **data** with sub-second inference **latency** for real-time sentiment analysis.

CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

EDUCATION

IMS Engineering College, Ghaziabad **September - 2020**

Bachelor of Technology